7. **(a)** A copper pipe has an internal diameter of 4.00 cm and an external diameter of 7.00 cm. The resistivity of copper is $1.68 \times 10^{-8}\ \Omega$ m.

Determine the resistance of 15.0 m of this pipe.

resistance = ... Ω [3]

(b) A number of resistors are connected to a 6.0 V cell in an electrical circuit as shown in Fig. 7.

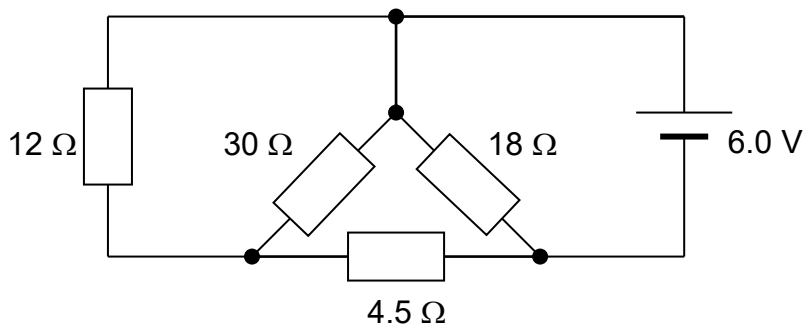

Fig. 7

With reference to Fig. 7,

(i) show that the equivalent resistance of the circuit is 7.6 Ω.

[1]

(ii) determine the current in the 18 Ω resistor.

current = ... A [1]

(iii) determine the power dissipated in the 4.5 Ω resistor.

power = ... W [3]